Module 2 Basic Content:

Drinking water sources

- Groundwater: a body of water that is naturally stored below ground in an aquifer; aquifers are either confined or unconfined
    - Odd factoid: most people tend to think of groundwater as water that is stored in underground pools or rivers that you can swim in, but groundwater storage is actually more like water stored in a soaked sponge – with soil, sand, and rock playing the part of the sponge
- Aquifer: an underground layer of soil, sand, or rock that is saturated with water
- Unconfined aquifer: an aquifer which does not have a “lid”, meaning there is NO layer of solid (impermeable) clay or rock over the top, so the water moves freely in and out of the aquifer
- Water table: a line at the top of an unconfined aquifer that marks the upper margin of the zone of saturation; this line moves as the water is added to (recharged) and removed from (discharged) the aquifer
    - Odd factoid: since they do not have a lid, unconfined aquifers – especially shallow ones – are much more likely to be influenced by both natural and human surface activity
- Confined aquifer: an aquifer which has a “lid”, meaning there is a layer of solid (impermeable) clay or rock over the top which contains the water and prevents it from moving outside the aquifer
- Artesian well: a free-flowing well that occurs when a confined aquifer is tapped and there is enough pressure in the aquifer to push the water up to the surface without a pump
- Surface water: a body of water that is naturally stored above ground; surface water is generally found as either a confined body of water (lakes) or a flowing watercourse (rivers and streams)
    - Odd factoid: In addition to providing drinking water, surface water sources are often used for multiple purposes such as transport, recreation, farming, ranching, industry, production of electricity, etc., and each of these uses has an impact on the water source

    - Odd factoid: Surface water is much more easily influenced by environmental conditions and human activity than groundwater

- Lake: body of surface water that is confined to a landform depression and filled by streams or runoff
    - Odd factoid: manmade lakes are often valleys with a stream of water flowing through them that are dammed on the lower end, so the lake fills with water
- River/stream: body of surface water that flows through a channel from points of higher elevation to points of lower elevation
- Springs/seeps: a natural transition point from groundwater to surface water where water moves from an aquifer to the land surface (springs produce more water than seeps); this usually occurs in places where heavy precipitation has raised the water table high enough that water flows free at the surface or where a stream channel has worn through rock to the water table

Drinking water systems

- Drinking water system: the organized means by which drinking water is collected from the source and distributed to the consumer for their consumption
    - Odd factoid: when the term “system” is used in water and wastewater, it often includes not only the physical parts of the system but also the people who work in and manage the system
    - Odd factoid: “consumers” are the people to whom drinking water systems and wastewater systems supply water and services
- Water source: the body of water used to provide drinking water; water sources can be groundwater, surface water, or springs
- Infrastructure: the physical components of the drinking water system; in a drinking water system, this mostly refers to the treatment facility, water lines/pipes, pumps, and water towers
    - Odd factoid: fire hydrants are part of the drinking water infrastructure and have two purposes: 1) points of quick access to water for the fire department, and 2) used to drain or flush water out of the lines quickly when repairing or maintaining water lines and towers

    - Odd factoid: the physical size of the fire hydrant is an indicator of the diameter of the water line under the hydrant; the larger the hydrant, the larger the water line – and the larger the volume of water it can produce at a moment's notice

- Water line/pipe: the underground tubes through which water is carried from water source to consumer
    - Odd factoid: Almost 2000 years ago, the Romans built some of the first water lines/pipes, called "aqueducts"; these are waterways built of stone or clay tile and used to carry water from sources in the mountains to towns and cities – and some of these aqueducts are still in use today
    - Odd factoid: PVC (polyvinyl chloride) water line is one of the more common types of water distribution line used in the US; it generally comes in sections which are 20 feet long (called sticks) and these sections are pieced together into water lines (which can be miles in length) using different fittings (e.g., collars, elbow joints, splitters) to make the line fit the terrain and needs of the consumer
    - Odd factoid: water line/pipe can be composed of many different materials (e.g., cast iron, steel, copper, PVC, concrete, clay tile, stone, lead) and come in a large range of diameters – with larger diameter pipes carrying many more gallons of water per minute than smaller diameter pipes
    - Odd factoid: since water lines/pipes are composed of so many different materials, the type of material used is often specific to the requirements for the situation, such as the temperature of the water being transported or the possibility that a water source may have naturally occurring compounds dissolved in it which can erode certain types of pipe; this is why the use of lead pipes for new water infrastructure was banned in the 1980's
        - Investigate the water crisis that occurred in Flint, Michigan to see how important this is
- Water tower (AKA standpipe): water storage facility in which water is pumped to hold water at a higher elevation
    - Odd factoid: in most places, water is distributed through water lines to consumers though passive means – meaning gravity is used to move

water from higher elevations to lower elevations; this is why water towers are almost always found on hills or built on a stand, and hold large volumes of water

    - Odd factoid: water stored in a water tower has high potential energy (energy of position) which is converted to kinetic energy by gravity as the water is released; the strong flow of water pressurizes the water system and distributes water through the lines to consumers

        - The taller the tower AND the more water it stores, the higher the potential energy and the higher the pressure it can produce – this is great because it means you can push water further down the line, *but there is a balance point to find* since most infrastructure is designed to withstand a certain amount of pressure and going over that can cause more damage than good

- Pump: a device used to transfer water from one place to another through pipes, often moving water across level ground or uphill against the pull of gravity; in a drinking water system, pumps are mainly used to move water from the water source to the water treatment facility and to fill water towers

- Drinking water treatment: various physical (e.g., filtration, reverse osmosis, ultraviolet/UV irradiation) and chemical (e.g., chlorine) methods used to manage source water quality, prior to distribution to consumers, by disinfecting the water (making it drinkable/potable) and/or adding fluoride

- Surface water treatment: surface water requires a much higher level of treatment due to the water being open to *contamination* from the environment and the number of *biologicals* living in the water

    - Odd factoid: environmental *contaminants* are compounds or sediments found in surface water that are usually introduced through runoff or human activity, which make the water undrinkable/non-potable; contaminants include things like soil, pesticides, herbicides, fertilizer, fuel/oil, chemicals released by industry, animal waste from feedlots, etc.

    - Odd factoid: *biologicals* range from microscopic bacteria and algae to things we can see, like invertebrates (e.g., crayfish and mussels) and vertebrates (e.g., fish and turtles)

- Groundwater treatment: *in general*, groundwater requires much less treatment compared to surface water because aquifers are relatively closed systems which are not easily influenced by surface activity; the closer the aquifer is to the surface, however, the more likely it is to be influenced by surface activity

- Wells: groundwater is pumped to the surface using wells, which are basically conduits from the surface directly into the aquifer; this makes wells one of the most likely places for groundwater to be contaminated through surface activity (e.g., flooding, cracks in well casing, over-pumping, illegal dumping)

    - Odd factoid: in shallow wells drilled into porous soil or rock, over-pumping can draw down the water table fast enough that liquids at or near the surface can be drawn down through the soil into the shallow aquifer; this is one way that chemicals applied to farm fields (e.g., pesticides, fertilizers, herbicides) can contaminate groundwater without actually washing into the well itself

Wastewater systems

- Wastewater: water that has been used and now carries contaminants which must be removed to return the water to a potable state so it is drinkable again

    - Odd factoid: like matter, water is neither created nor destroyed – the water we have today has always been here because of the water cycle which maintains a continuous cycle of use and replenishment; yes, we are drinking water that many, many, many millions of cycles ago was dinosaur urine

- Effluent: cleaned wastewater that is piped out of a wastewater treatment plant and returned to a local body of water; the wastewater has been treated/disinfected to a point that is considered safe to be released in the environment

- Wastewater system: the organized means by which wastewater is collected, treated, and returned to the environment; wastewater systems can be centralized or decentralized

    - Odd factoid: in the water cycle, infiltration and percolation are the two transport processes which work as nature's wastewater system through filtration and adsorption; when humans congregate in the same place in large numbers, though, this natural system is easily overwhelmed so human wastewater systems use the processes found

in nature (i.e., filtration and adsorption) and other treatments to clean and disinfect water in a much more intensive way

- Centralized wastewater system: sewage system for larger populations wherein wastewater is collected across a populated area and delivered to a central wastewater treatment facility; centralized systems are found in urban areas with higher population density
- Decentralized (onsite) wastewater system: wastewater systems for single households (or very small communities) wherein wastewater is collected and treated onsite through the use of a septic tank, lateral lines, aeration system, or lagoons; decentralized systems are predominantly found in rural areas with low population density
- Infrastructure: the physical components of the wastewater system, used for collection and treatment; in a wastewater system, this mostly refers to wastewater lines/pipes, septic tanks, sanitary sewers (pipes and tunnels), aerators, pumps/lift stations, lagoons, and the treatment facility
    - Odd factoid: similar to water systems, wastewater is moved using gravity so most pipes have a slight tilt from one end to the other to keep things flowing
- Wastewater lines/pipes: tubes through which wastewater is collected from various household sources (e.g., sink, toilet, washing machine) and carried away for treatment; in rural areas it will be delivered to the household septic tank – in urban areas it will be delivered to a sanitary sewer system leading to the wastewater treatment plant (used in centralized and decentralized systems)
- Pump/lift station: a device used to transfer wastewater from one place to another through pipes, often moving wastewater across level ground or uphill against the pull of gravity; in a wastewater system, a lift station is a specific type of pump used mainly to move wastewater uphill when the local terrain is hilly or the wastewater collection area is at a lower elevation than the treatment facility (used in centralized systems)
- Septic tank: collection and holding tank for local household wastewater wherein bacteria break down solid human waste (e.g., paper, fecal matter) and the effluent (liquid portion) is piped to a leach field (used in decentralized systems)

    - Odd factoid: two reasons for *not* putting just anything down the sink or toilet: 1) the flow in wastewater lines is based on gravity so it does not take much to stop the flow (get the plunger or if it's bad enough, call a plumber), and 2) bacteria will not breakdown inorganic products (e.g., plastic, toys), so those collect in the septic tank and fill it up

- Sanitary sewer system: larger lines/pipes which are used to collect wastewater from multiple households and businesses and deliver it to the wastewater treatment facility (used in centralized systems)
    - Odd factoid: the closer the wastewater lines/pipes of the sewer are to the treatment facility, the larger their diameter because they must carry a larger volume of wastewater
- Stormwater system: larger lines/pipes which are used to collect rain water or snow melt from streets and deliver these potentially large volumes of water to a local lake or stream where it is released untreated (used in centralized systems)
    - Odd factoid: Since stormwater sewer systems release *untreated* water into local surface water sources, it is illegal to dump things that can pollute water into these drains (e.g., used motor oil, chemicals, paint)
- Leach field/lateral lines: an area of land (often in the side yard or backyard of a house) where perforated pipes called lateral lines are buried and used to distribute the effluent that leaves a septic tank; the leach field is spread out over a larger area to ensure that: 1) the wastewater drains quickly (percolates), 2) there is a large enough volume of soil to filter and adsorb the waste products dissolved in the water, so 3) the waste products dissolved in the water do not accumulate in a specific area and become a hazard (used in decentralized systems)
    - Odd factoid: if there is a place in your yard where the grass grows very fast in the summer, this is likely the leach field for your septic system; the grass grows fast because not only is it getting more water than the rest of your yard, it is also getting nutrients from the wastewater
    - Odd factoid: the lateral lines buried in a leach field are relatively shallow, so the placement of the leach field is important (this is why they are often found in yards); cars, trucks, and heavy equipment should not be driven over the leach field because it is possible to break the lateral lines and collapse the field

- Aerators: a system in which air is pumped through a tank holding wastewater to increase the speed with which the bacteria breakdown the organic matter; aerators are often added into other systems (septic tanks, leach fields, lagoons) to improve their performance (used in centralized and decentralized systems)
- Lagoons: pond-like structures which hold the effluent that flows from a septic tank; increased access to air, UV rays from sunlight, and evaporation due to sun and wind finish disinfecting the wastewater (used in centralized and decentralized systems)
    - Odd factoid: lagoons are designed to work well with a certain amount of rainfall, so times of drought and times of excess rain have a negative effect on the proper functioning of lagoons
- Wastewater treatment facility: a central place where wastewater is collected and treated/disinfected; as wastewater is moved from tank to tank in the facility, different methods of treatment are performed in each tank (e.g., filtration, settling, disinfection) and the water becomes cleaner with each method until the effluent is eventually clean enough to be released to a body of water (used in centralized systems)